

Azure Managed Lustre

MLPerf® Storage v3.0 Submission

System Description, Configuration & Tuning Report

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1. Solution Overview

Azure Managed Lustre is a fully managed, cloud-native parallel file system service that delivers the open-source Lustre® file system as a first-party Microsoft Azure resource. It is purpose-built for high-performance computing (HPC) and AI/ML workloads that require high aggregate throughput and a POSIX-compliant shared namespace across large fleets of compute clients.

For this MLPerf Storage v3.0 submission, Azure Managed Lustre was evaluated across a range of performance tiers and capacities. The results demonstrate the throughput scaling of the service for the UNet3D training data pipeline and for large-language-model (llama3) checkpoint read/write workloads.

1.1 Fully Managed Service

Azure Managed Lustre removes the operational burden of deploying and running a Lustre cluster. Microsoft provisions, configures, monitors, patches, and maintains the underlying Lustre server components, the Management Server (MGS), Metadata Servers (MDS) with their Metadata Targets (MDT), and Object Storage Servers (OSS) with their Object Storage Targets (OST). Customers simply select a performance tier and a capacity, and the file system is created in minutes with a single endpoint.

- No server management: MDS/MDT and OSS/OST are deployed and operated by Azure.
- Durable, SSD-backed storage with data resilience that survives node loss.
- Optional Azure Blob Storage integration (Hierarchical Storage Management) for import/export of data.
- Built-in monitoring, security and encryption at rest.

1.2 Global Availability

Azure Managed Lustre is available globally across a broad set of Microsoft Azure regions, allowing customers to co-locate the file system with their compute in the region closest to their workload. This submission itself was executed across multiple Azure regions (for example US, Europe and Asia-Pacific regions), underscoring the consistency of the service across geographies.

1.3 Direct Virtual Network Integration

A defining architectural characteristic of Azure Managed Lustre is that the Lustre server components are injected directly into the customer's Azure Virtual Network. The network interface cards (NICs) of the MGS, MDS and OSS nodes are attached to a customer-designated subnet, so Lustre traffic (LNet over TCP) flows natively inside the customer Azure Virtual Network rather than traversing a gateway or proxy.

- The file-system servers present real NICs with private IP addresses in the customer subnet.
- Clients mount the file system over the native Lustre network (LNet/TCP) using the server NID (e.g. 10.0.x.x@tcp), exactly as seen in the benchmark job scripts.
- No public endpoints are required; traffic stays within the Azure Virtual Network security boundary.
- Throughput scales with the number of OSS provisioned, with a function of the selected tier and capacity.

The following diagrams illustrate the architecture. Figure 1 shows how the fully managed file system injects its server NICs directly into the customer virtual network, alongside the CPU clients.

Azure Managed Lustre — Virtual Network Injection

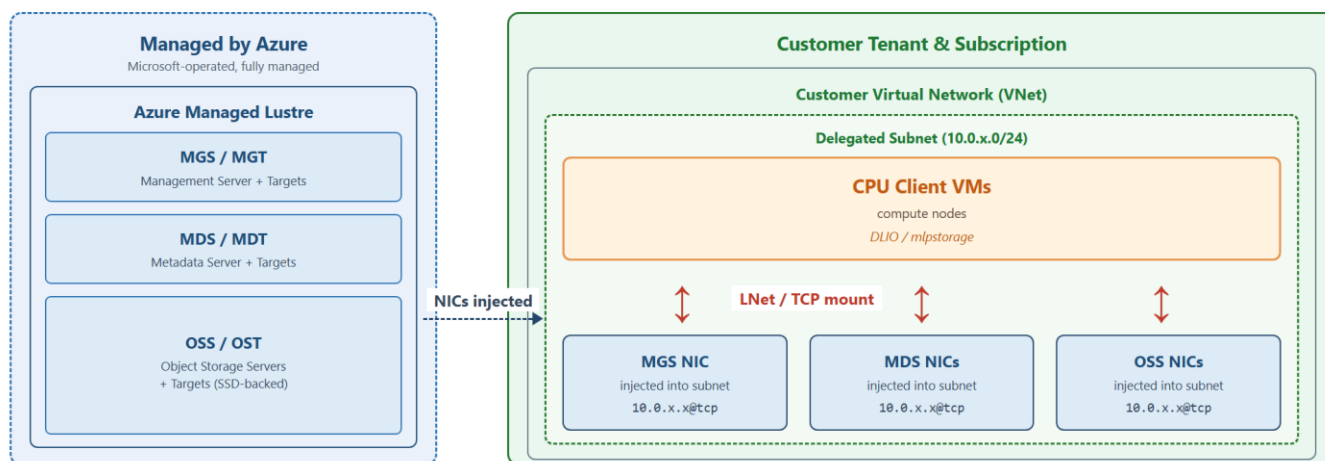


Figure 1, Azure Managed Lustre server topology. The Azure Managed Lustre instances are fully managed by Microsoft and the NICs enabling the filesystem mount are exposed in customer Azure Virtual Network.

These benchmarks have been executed inside a series of Slurm clusters in different Azure regions managed by Azure CycleCloud. The whole scheme is visible in Figure 2 adds the orchestration layer, in which Azure CycleCloud provisions a Slurm cluster whose compute nodes are used in this benchmark run as the Lustre clients. Azure CycleCloud allows to scale elastically a Slurm cluster in Azure based on the job demand

Azure Managed Lustre with Azure CycleCloud + Slurm Orchestration

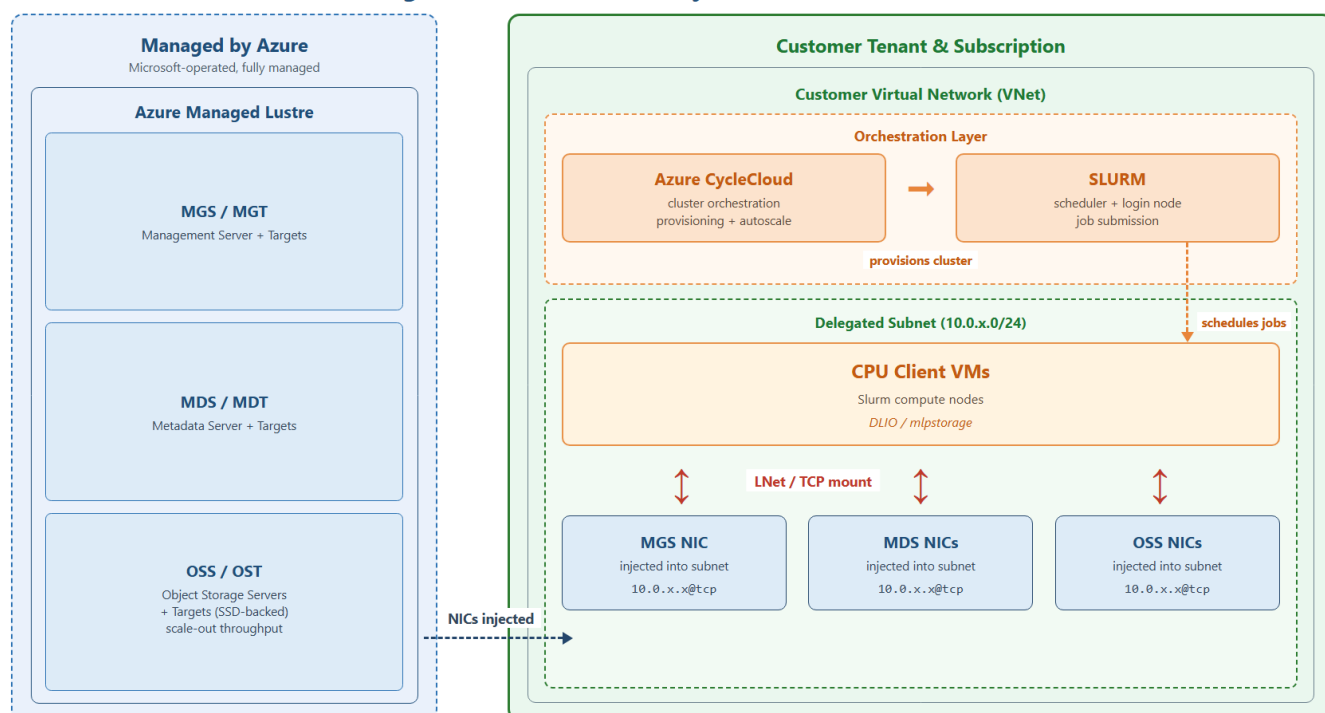


Figure 2, Azure CycleCloud allows customers to easily provision and orchestrate Slurm clusters, ready to run large scale AI and HPC workloads distributed across GPU and CPU specialized nodes.

2. Performance Tiers

Azure Managed Lustre offers multiple durable, SSD-backed performance tiers. The tiers differ primarily in their capacity density, the amount of usable storage bundled into each provisioning increment, rather than in the bandwidth of that increment.

In every tier, each capacity increment delivers approximately 2 GB/s of throughput; the tiers simply package that same unit of bandwidth with different capacity. A higher throughput-per-TiB tier therefore reaches the same aggregate bandwidth with less provisioned capacity, while a lower throughput-per-TiB tier bundles more capacity behind the same bandwidth unit.

SKU (tier)	Throughput per TiB	Storage minimum	Capacity increment	Bandwidth per increment
AMLFS-Durable-Premium-20	20 MB/s	96 TiB	96 TiB	2 GB/s
AMLFS-Durable-Premium-40	40 MB/s	48 TiB	48 TiB	2 GB/s
AMLFS-Durable-Premium-125	125 MB/s	16 TiB	16 TiB	2 GB/s
AMLFS-Durable-Premium-250	250 MB/s	8 TiB	8 TiB	2 GB/s
AMLFS-Durable-Premium-500	500 MB/s	4 TiB	4 TiB	2 GB/s

Each increment provides roughly the same 2 GB/s of bandwidth (throughput-per-TiB × increment size ≈ 2 GB/s across the 20, 40, 125, 250, and 500 tiers): the tiers trade capacity density against throughput density, not raw per-increment bandwidth.

The UNet3D training runs in this submission are designed to demonstrate that the different tiers can deliver the same Accelerator Utilization (AU) and the same throughput to the training workload, differing only in the provisioned capacity required to do so. Because each increment carries ~2 GB/s regardless of tier, a high-density tier (e.g. 500 MB/s/TiB) and a low-density tier (e.g. 20 MB/s/TiB) can both saturate the same client throughput and reach the same AU, the high-throughput-per-TiB tiers simply do it with far less capacity. The per-system results therefore show equivalent AU/throughput at markedly different capacity points.

The same principle lets the file system be right sized for checkpointing. Because each capacity increment carries ~2 GB/s, an AMLFS-125 file system can be provisioned at the exact size whose aggregate bandwidth matches the bandwidth each checkpoint requires, so the checkpoint write and read time, and therefore the GPU utilization during checkpointing, is kept at the required target. In this submission the checkpointing workloads all use the AMLFS-125 tier at different sizes (for example 160 TiB for llama3-8b, 1,280 TiB for llama3-70b, and 4,096 TiB for the larger llama3-405b and llama3-1t models), scaling capacity, and with it bandwidth, to the checkpoint size of each model.

3. Benchmark Clients

Benchmarks were driven by standard Azure virtual machines running unmodified, installed kernel-module open-source Lustre clients on Ubuntu 24.04. Three client VM families were used, matching the workload. All client hardware specifications below are taken from the official Microsoft Learn VM size documentation.

Client VM SKU	Series	vCPUs	Memory	Max NICs	Max network	Used for
Standard_E192is_v6	Esv6-series (memory optimized)	192	1,832 GiB	8	200,000 Mbps (200 Gbps)	Checkpointing (llama3)
Standard_D192ds_v6	Ddsv6-series (general purpose)	192	768 GiB	8	82,000 Mbps (82 Gbps)	Training (UNet3D)
Standard_E104is_v5	Ev5-series (memory optimized)	104	672 GiB	8	100,000 Mbps (100 Gbps)	Checkpointing (llama3)

All three SKUs are Intel x86-64 platforms exposed to the guest with hyper-threading enabled. Clients mount the file system over LNet/TCP.

Standard_E192is_v6, full specification

Attribute	Value
VM series	Esv6-series (memory optimized)
Processor	Intel Xeon Platinum 8573C (Emerald Rapids)

Attribute	Value
Clock	3.0 GHz all-core turbo
vCPUs	192
Memory	1,832 GB
Local (temp) storage	None (remote managed disks only)
Max NICs	8
Max network bandwidth	200,000 Mbps (200 Gbps)
OS image	Azure HPC Ubuntu 24.04 (microsoft-dsvm:ubuntu-hpc:2404:24.04.2026052601)
Kernel	6.8.0-1054-azure
Lustre client	2.16.1_14_gbc76088

Source: <https://learn.microsoft.com/en-us/azure/virtual-machines/sizes/memory-optimized/esv6-series>

Standard_D192ds_v6, full specification

Attribute	Value
VM series	Ddsv6-series (general purpose)
Processor	Intel Xeon Platinum 8573C (Emerald Rapids)
Clock	3.0 GHz all-core turbo
vCPUs	192
Memory	768 GiB
Local (temp) storage	Up to 10,560 GiB NVMe (6 × 1,760 GiB temp disks)
Max NICs	8
Max network bandwidth	82,000 Mbps (82 Gbps)
OS image	Azure HPC Ubuntu 24.04 (microsoft-dsvm:ubuntu-hpc:2404:24.04.2026052601)
Kernel	6.8.0-1054-azure
Lustre client	2.16.1_14_gbc76088

Source: <https://learn.microsoft.com/en-us/azure/virtual-machines/sizes/general-purpose/ddsv6-series>

Standard_E104is_v5, full specification

Attribute	Value
VM series	Ev5-series (memory optimized)
Processor	Intel Xeon Platinum 8573C (Emerald Rapids) / 8473C (Sapphire Rapids) / 8370C (Ice Lake)
Clock	3.5 GHz all-core turbo
vCPUs	104
Memory	672 GiB
Local (temp) storage	None (remote managed disks only)
Max NICs	8
Max network bandwidth	100,000 Mbps (100 Gbps)
OS image	Azure HPC Ubuntu 24.04 (microsoft-dsvm:ubuntu-hpc:2404:24.04.2026052601)
Kernel	6.8.0-1054-azure
Lustre client	2.16.1_14_gbc76088

Source: <https://learn.microsoft.com/en-us/azure/virtual-machines/sizes/memory-optimized/ev5-series>

Client Software & OS Image

All clients ran the standard Azure HPC Ubuntu 24.04 marketplace image, which ships a Microsoft-maintained, pre-integrated Lustre client. No custom kernel or client build was required, the Lustre client is installed as kernel module inside the Azure HPC image.

Component	Value
Marketplace image (URN)	microsoft-dsvm:ubuntu-hpc:2404:24.04.2026052601
Operating system	Ubuntu 24.04 LTS (Azure HPC image)
Kernel	6.8.0-1054-azure (#60-Ubuntu SMP Wed Apr 15 16:02:04 UTC 2026, x86_64)
Lustre client version	2.16.1_14_gbc76088

The Lustre client used across all systems has the following characteristics:

- The Azure HPC Ubuntu image bundles the Lustre client that is matched and validated against the Azure Managed Lustre server version.
- Lustre client packages are also published in the Microsoft package repositories, so they can be installed on other supported Ubuntu images.
- The Lustre client source code used by Azure Managed Lustre is publicly available at github.com/microsoft/amlFilesystem-lustre.

4. Systems Under Test

Sixteen system configurations were submitted, spanning five throughput tiers and a range of capacities and client counts. Each configuration is named AMLFS-Durable-Premium-<tier>-<capacity>-<clients>-<client VM>.

System	Tier	Capacity	Clients	Client VM	Workload
AMLFS-Durable-Premium-125-1280TiB-16-Standard_E192is_v6	125 MB/s/TiB	1280 TiB	16	Standard_E192is_v6	Checkpointing
AMLFS-Durable-Premium-125-160TiB-2-Standard_E192is_v6	125 MB/s/TiB	160 TiB	2	Standard_E192is_v6	Checkpointing
AMLFS-Durable-Premium-125-160TiB-3-Standard_D192ds_v6	125 MB/s/TiB	160 TiB	3	Standard_D192ds_v6	UNet3D training
AMLFS-Durable-Premium-125-2400TiB-46-Standard_D192ds_v6	125 MB/s/TiB	2400 TiB	46	Standard_D192ds_v6	UNet3D training
AMLFS-Durable-Premium-125-4096TiB-128-Standard_E104is_v5	125 MB/s/TiB	4096 TiB	128	Standard_E104is_v5	Checkpointing
AMLFS-Durable-Premium-125-4096TiB-70-Standard_D192ds_v6	125 MB/s/TiB	4096 TiB	70	Standard_D192ds_v6	UNet3D training
AMLFS-Durable-Premium-20-25632TiB-70-Standard_D192ds_v6	20 MB/s/TiB	25632 TiB	70	Standard_D192ds_v6	UNet3D training
AMLFS-Durable-Premium-125-800TiB-16-Standard_D192ds_v6	125 MB/s/TiB	800 TiB	16	Standard_D192ds_v6	UNet3D training
AMLFS-Durable-Premium-20-4800TiB-16-Standard_D192ds_v6	20 MB/s/TiB	4800 TiB	16	Standard_D192ds_v6	UNet3D training
AMLFS-Durable-Premium-20-960TiB-3-Standard_D192ds_v6	20 MB/s/TiB	960 TiB	3	Standard_D192ds_v6	UNet3D training
AMLFS-Durable-Premium-250-400TiB-16-Standard_D192ds_v6	250 MB/s/TiB	400 TiB	16	Standard_D192ds_v6	UNet3D training
AMLFS-Durable-Premium-250-80TiB-3-Standard_D192ds_v6	250 MB/s/TiB	80 TiB	3	Standard_D192ds_v6	UNet3D training
AMLFS-Durable-Premium-40-2400TiB-16-Standard_D192ds_v6	40 MB/s/TiB	2400 TiB	16	Standard_D192ds_v6	UNet3D training
AMLFS-Durable-Premium-40-480TiB-3-Standard_D192ds_v6	40 MB/s/TiB	480 TiB	3	Standard_D192ds_v6	UNet3D training
AMLFS-Durable-Premium-500-200TiB-16-Standard_D192ds_v6	500 MB/s/TiB	200 TiB	16	Standard_D192ds_v6	UNet3D training
AMLFS-Durable-Premium-500-40TiB-3-Standard_D192ds_v6	500 MB/s/TiB	40 TiB	3	Standard_D192ds_v6	UNet3D training

5. Per-System Configuration & Tunings

The following sections detail, for every submitted run, the clients used, the Lustre mount, the client-side tuning parameters applied, the file-system striping, and the exact benchmark invocation.

5.1 AMLFS-Durable-Premium-125-1280TiB-16-Standard_E192is_v6

- Tier / capacity: 125 MB/s/TiB, 1280 TiB provisioned
- Clients: 16 × Standard_E192is_v6 (192 vCPU, 1,832 GiB RAM per SKU; Azure HPC Ubuntu 24.04, Lustre client 2.16.1)

Checkpointing, llama3-70b

Client tunings (lctl set_param)

```
llite.*.hybrid_io=1
llite.*.hybrid_io_read_threshold_bytes=1073741824
mdc.*.max_mod_rpcs_in_flight=96
osc.*.max_rpcs_in_flight=256
llite.*.max_read_ahead_mb=8192
llite.*.max_read_ahead_per_file_mb=1024
llite.*.max_cached_mb=128000
```

Striping

```
lfs setstripe -c 5 -S 1M /mlperf-storage/mlpstorage_run
```

Cache handling

Page cache + DLM LRU flushed between phases (cold-cache reads).

Command

mlpstorage command	Flags
closed checkpointing run file	--model llama3-70b --num-processes 64 --client-host-memory-in-gb 1805 --num-checkpoints-read 0 --num-checkpoints-write 10
closed checkpointing run file	--model llama3-70b --num-processes 64 --client-host-memory-in-gb 1805 --num-checkpoints-read 10 --num-checkpoints-write 0

Run timestamps: 20260722_185605, 20260722_185939

5.2 AMLFS-Durable-Premium-125-160TiB-2-Standard_E192is_v6

- Tier / capacity: 125 MB/s/TiB, 160 TiB provisioned
- Clients: 2 × Standard_E192is_v6 (192 vCPU, 1,832 GiB RAM per SKU; Azure HPC Ubuntu 24.04, Lustre client 2.16.1)

Checkpointing, llama3-8b

Client tunings (lctl set_param)

```
llite.*.hybrid_io=1
llite.*.hybrid_io_read_threshold_bytes=1073741824
mdc.*.max_mod_rpcs_in_flight=96
osc.*.max_rpcs_in_flight=256
llite.*.max_read_ahead_mb=8192
llite.*.max_read_ahead_per_file_mb=1024
llite.*.max_cached_mb=128000
```

Striping

```
lfs setstripe -c 4 -S 1M /mlperf-storage/mlpstorage_run
```

Cache handling: page cache + DLM LRU flushed between phases (cold-cache reads).

Command

mlpstorage command	Flags
closed checkpointing run file	--model llama3-8b --num-processes 8 --client-host-memory-in-gb 1805 --num-checkpoints-read 0 --num-checkpoints-write 10
closed checkpointing run file	--model llama3-8b --num-processes 8 --client-host-memory-in-gb 1805 --num-checkpoints-read 10 --num-checkpoints-write 0

Run timestamps: 20260722_193256, 20260722_193609

5.3 AMLFS-Durable-Premium-125-160TiB-3-Standard_D192ds_v6

- Tier / capacity: 125 MB/s/TiB, 160 TiB provisioned
- Clients: 3 × Standard_D192ds_v6 (192 vCPU, 768 GiB RAM per SKU; Azure HPC Ubuntu 24.04, Lustre client 2.16.1)

Training, UNet3D

Client tunings (lctl set_param)

```
mdc.*.max_mod_rpcs_in_flight=96
osc.*.max_rpcs_in_flight=128
llite.*.max_read_ahead_mb=2048
llite.*.max_read_ahead_per_file_mb=512
llite.*.max_read_ahead_whole_mb=256
```

Striping

```
lfs setstripe -c -1 -S 4M /mlperf-storage
```

Command

mlpstorage command	Flags	DLIO params
closed training unet3d run file	--num-accelerators 3 --accelerator-type b200 --client-host-memory-in-gb 768 --num-client-hosts 3	reader.odirect=True reader.read_threads=96 dataset.num_files_train=84375

Run timestamps: 20260721_210532, 20260721_220535, 20260721_230538, 20260722_000542, 20260722_010544, 20260722_020547

5.4 AMLFS-Durable-Premium-125-2400TiB-46-Standard_D192ds_v6

- Tier / capacity: 125 MB/s/TiB, 2400 TiB provisioned
- Clients: 46 × Standard_D192ds_v6 (192 vCPU, 768 GiB RAM per SKU; Azure HPC Ubuntu 24.04, Lustre client 2.16.1)

Training, UNet3D

Client tunings (lctl set_param)

```
mdc.*.max_mod_rpcs_in_flight=96
osc.*.max_rpcs_in_flight=128
llite.*.max_read_ahead_mb=2048
llite.*.max_read_ahead_per_file_mb=512
llite.*.max_read_ahead_whole_mb=256
```

Striping

```
lfs setstripe -c 10 -S 4M /mlperf-storage
```

Command

mlpstorage command	Flags	DLIO params
closed training unet3d run file	--num-accelerators 46 --accelerator-type b200 --client-host-memory-in-gb 768 --num-client-hosts 46	reader.odirect=True reader.read_threads=64 dataset.num_files_train=1293758

Run timestamps: 20260722_084136, 20260722_094540, 20260722_104946, 20260722_115350, 20260722_125756, 20260722_140202

5.5 AMLFS-Durable-Premium-125-4096TiB-128-Standard_E104is_v5

- Tier / capacity: 125 MB/s/TiB, 4096 TiB provisioned
- Clients: 128 × Standard_E104is_v5 (104 vCPU, 672 GiB RAM per SKU; Azure HPC Ubuntu 24.04, Lustre client 2.16.1)

Checkpointing, llama3-1t

Client tunings (lctl set_param)

```
mdc.*.max_mod_rpcs_in_flight=96
osc.*.max_rpcs_in_flight=128
llite.*.max_read_ahead_mb=8192
llite.*.max_read_ahead_per_file_mb=1024
llite.*.max_cached_mb=32000
```

Striping

```
lfs setstripe -c 1 -S 4M /mlperf-storage/mlpstorage_run
```

Cache handling

Page cache + DLM LRU flushed between phases (cold-cache reads).

Command

mlpstorage command	Flags	DLIO params
closed checkpointing run file	--model llama3-1t --num-processes 1024 --client-host-memory-in-gb 670 --num-checkpoints-read 0 --num-checkpoints-write 10	—
closed checkpointing run file	--model llama3-1t --num-processes 1024 --client-host-memory-in-gb 670 --num-checkpoints-read 10 --num-checkpoints-write 0	—

Run timestamps: 20260714_164815, 20260714_170300

Checkpointing, llama3-405b

Client tunings (lctl set_param)

```
llite.*.hybrid_io=1
llite.*.hybrid_io_read_threshold_bytes=1073741824
mdc.*.max_mod_rpcs_in_flight=96
osc.*.max_rpcs_in_flight=256
llite.*.max_read_ahead_mb=8192
llite.*.max_read_ahead_per_file_mb=1024
llite.*.max_cached_mb=32000
```

Striping

```
lfs setstripe -c 4 -S 1M /mlperf-storage/mlpstorage_run
```

Cache handling

Page cache + DLM LRU flushed between phases (cold-cache reads).

Command

mlpstorage command	Flags	DLIO params
closed checkpointing run file	--model llama3-405b --num-processes 512 --client-host-memory-in-gb 670 --num-checkpoints-read 0 --num-checkpoints-write 10	—
closed checkpointing run file	--model llama3-405b --num-processes 512 --client-host-memory-in-gb 670 --num-checkpoints-read 10 --num-checkpoints-write 0	—

5.6 AMLFS-Durable-Premium-125-4096TiB-70-Standard_D192ds_v6

- Tier / capacity: 125 MB/s/TiB, 4096 TiB provisioned
- Clients: 70 × Standard_D192ds_v6 (192 vCPU, 768 GiB RAM per SKU; Azure HPC Ubuntu 24.04, Lustre client 2.16.1)

Training, UNet3D

Client tunings (lctl set_param)

```
mdc.*.max_mod_rpcs_in_flight=96
osc.*.max_rpcs_in_flight=128
llite.*.max_read_ahead_mb=2048
llite.*.max_read_ahead_per_file_mb=512
llite.*.max_read_ahead_whole_mb=256
```

Striping

```
lfs setstripe -c 10 -S 4M /mlperf-storage
```

Command

mlpstorage command	Flags	DLIO params
closed training unet3d run file	--num-accelerators 70 --accelerator-type b200 --client-host-memory-in-gb 768 --num-client-hosts 70	reader.odirect=True reader.read_threads=96 dataset.num_files_train=1968762

Run timestamps: 20260722_215413, 20260722_230045, 20260723_000650, 20260723_011257, 20260723_021900, 20260723_032605

5.7 AMLFS-Durable-Premium-125-800TiB-16-Standard_D192ds_v6

- Tier / capacity: 125 MB/s/TiB, 800 TiB provisioned
- Clients: 16 × Standard_D192ds_v6 (192 vCPU, 768 GiB RAM per SKU; Azure HPC Ubuntu 24.04, Lustre client 2.16.1)

Training, UNet3D

Client tunings (lctl set_param)

```
mdc.*.max_mod_rpcs_in_flight=96
osc.*.max_rpcs_in_flight=128
llite.*.max_read_ahead_mb=2048
llite.*.max_read_ahead_per_file_mb=512
llite.*.max_read_ahead_whole_mb=256
```

Striping

```
lfs setstripe -c -1 -S 4M /mlperf-storage
```

Command

mlpstorage command	Flags	DLIO params
closed training unet3d run file	--num-accelerators 16 --accelerator-type b200 --client-host-memory-in-gb 768 --num-client-hosts 16	reader.odirect=True reader.read_threads=96 dataset.num_files_train=450002

Run timestamps: 20260723_014923, 20260723_025226, 20260723_035629, 20260723_050135, 20260723_060438, 20260723_070742

5.8 AMLFS-Durable-Premium-20-4800TiB-16-Standard_D192ds_v6

- Tier / capacity: 20 MB/s/TiB, 4800 TiB provisioned

- Clients: 16 × Standard_D192ds_v6 (192 vCPU, 768 GiB RAM per SKU; Azure HPC Ubuntu 24.04, Lustre client 2.16.1)

Training, UNet3D

Client tunings (lctl set_param)

```
mdc.*.max_mod_rpcs_in_flight=96
osc.*.max_rpcs_in_flight=128
llite.*.max_read_ahead_mb=2048
llite.*.max_read_ahead_per_file_mb=512
llite.*.max_read_ahead_whole_mb=256
```

Striping

```
lfs setstripe -c -1 -S 4M /mlperf-storage
```

Command

mlpstorage command	Flags	DLIO params
closed training unet3d run file	--num-accelerators 16 --accelerator-type b200 --client-host-memory-in-gb 768 --num-client-hosts 16	reader.odirect=True reader.read_threads=96 dataset.num_files_train=450002

Run timestamps: 20260722_064421, 20260722_074626, 20260722_084928, 20260722_095231, 20260722_105534, 20260722_115837

5.9 AMLFS-Durable-Premium-20-960TiB-3-Standard_D192ds_v6

- Tier / capacity: 20 MB/s/TiB, 960 TiB provisioned
- Clients: 3 × Standard_D192ds_v6 (192 vCPU, 768 GiB RAM per SKU; Azure HPC Ubuntu 24.04, Lustre client 2.16.1)

Training, UNet3D

Client tunings (lctl set_param)

```
mdc.*.max_mod_rpcs_in_flight=96
osc.*.max_rpcs_in_flight=128
llite.*.max_read_ahead_mb=2048
llite.*.max_read_ahead_per_file_mb=512
llite.*.max_read_ahead_whole_mb=256
```

Striping

```
lfs setstripe -c -1 -S 4M /mlperf-storage
```

Command

mlpstorage command	Flags	DLIO params
closed training unet3d run file	--num-accelerators 3 --accelerator-type b200 --client-host-memory-in-gb 768 --num-client-hosts 3	reader.odirect=True reader.read_threads=96 dataset.num_files_train=84375

Run timestamps: 20260721_210630, 20260721_220634, 20260721_230636, 20260722_000639, 20260722_010641, 20260722_020643

5.10 AMLFS-Durable-Premium-250-400TiB-16-Standard_D192ds_v6

- Tier / capacity: 250 MB/s/TiB, 400 TiB provisioned
- Clients: 16 × Standard_D192ds_v6 (192 vCPU, 768 GiB RAM per SKU; Azure HPC Ubuntu 24.04, Lustre client 2.16.1)

Training, UNet3D

Client tunings (lctl set_param)

```
mdc.*.max_mod_rpc_in_flight=96
osc.*.max_rpc_in_flight=128
llite.*.max_read_ahead_mb=2048
llite.*.max_read_ahead_per_file_mb=512
llite.*.max_read_ahead_whole_mb=256
```

Striping

```
lfs setstripe -c -1 -S 4M /mlperf-storage
```

Command

mlpstorage command	Flags	DLIO params
closed training unet3d run file	--num-accelerators 16 --accelerator-type b200 --client-host-memory-in-gb 768 --num-client-hosts 16	reader.odirect=True reader.read_threads=96 dataset.num_files_train=450002

Run timestamps: 20260723_121614, 20260723_131917, 20260723_142221, 20260723_152525, 20260723_162827, 20260723_173130

5.11 AMLFS-Durable-Premium-250-80TiB-3-Standard_D192ds_v6

- Tier / capacity: 250 MB/s/TiB, 80 TiB provisioned
- Clients: 3 × Standard_D192ds_v6 (192 vCPU, 768 GiB RAM per SKU; Azure HPC Ubuntu 24.04, Lustre client 2.16.1)

Training, UNet3D

Client tunings (lctl set_param)

```
mdc.*.max_mod_rpc_in_flight=96
osc.*.max_rpc_in_flight=128
llite.*.max_read_ahead_mb=2048
llite.*.max_read_ahead_per_file_mb=512
llite.*.max_read_ahead_whole_mb=256
```

Striping

```
lfs setstripe -c -1 -S 4M /mlperf-storage
```

Command

mlpstorage command	Flags	DLIO params
closed training unet3d run file	--num-accelerators 3 --accelerator-type b200 --client-host-memory-in-gb 768 --num-client-hosts 3	reader.odirect=True reader.read_threads=96 dataset.num_files_train=84375

Run timestamps: 20260721_160519, 20260721_170521, 20260721_180525, 20260721_190527, 20260721_200529, 20260721_210532

5.12 AMLFS-Durable-Premium-40-2400TiB-16-Standard_D192ds_v6

- Tier / capacity: 40 MB/s/TiB, 2400 TiB provisioned
- Clients: 16 × Standard_D192ds_v6 (192 vCPU, 768 GiB RAM per SKU; Azure HPC Ubuntu 24.04, Lustre client 2.16.1)

Training, UNet3D

Client tunings (lctl set_param)

```
mdc.*.max_mod_rpc_in_flight=96
osc.*.max_rpc_in_flight=128
llite.*.max_read_ahead_mb=2048
llite.*.max_read_ahead_per_file_mb=512
llite.*.max_read_ahead_whole_mb=256
```

Striping

```
lfs setstripe -c -1 -S 4M /mlperf-storage
```

Command

mlpstorage command	Flags	DLIO params
closed training unet3d run file	--num-accelerators 16 --accelerator-type b200 --client-host-memory-in-gb 768 --num-client-hosts 16	reader.odirect=True reader.read_threads=96 dataset.num_files_train=450002

Run timestamps: 20260722_151607, 20260722_161911, 20260722_172214, 20260722_182517, 20260722_193122, 20260722_203524

5.13 AMLFS-Durable-Premium-40-480TiB-3-Standard_D192ds_v6

- Tier / capacity: 40 MB/s/TiB, 480 TiB provisioned
- Clients: 3 × Standard_D192ds_v6 (192 vCPU, 768 GiB RAM per SKU; Azure HPC Ubuntu 24.04, Lustre client 2.16.1)

Training, UNet3D

Client tunings (lctl set_param)

```
mdc.*.max_mod_rpcs_in_flight=96  
osc.*.max_rpcs_in_flight=128  
llite.*.max_read_ahead_mb=2048  
llite.*.max_read_ahead_per_file_mb=512  
llite.*.max_read_ahead_whole_mb=256
```

Striping

```
lfs setstripe -c -1 -S 4M /mlperf-storage
```

Command

mlpstorage command	Flags	DLIO params
closed training unet3d run file	--num-accelerators 3 --accelerator-type b200 --client-host-memory-in-gb 768 --num-client-hosts 3	reader.odirect=True reader.read_threads=96 dataset.num_files_train=84375

Run timestamps: 20260721_155950, 20260721_170053, 20260721_180156, 20260721_190300, 20260721_200404, 20260721_210506

5.14 AMLFS-Durable-Premium-500-200TiB-16-Standard_D192ds_v6

- Tier / capacity: 500 MB/s/TiB, 200 TiB provisioned
- Clients: 16 × Standard_D192ds_v6 (192 vCPU, 768 GiB RAM per SKU; Azure HPC Ubuntu 24.04, Lustre client 2.16.1)

Training, UNet3D

Client tunings (lctl set_param)

```
mdc.*.max_mod_rpcs_in_flight=96  
osc.*.max_rpcs_in_flight=128  
llite.*.max_read_ahead_mb=2048  
llite.*.max_read_ahead_per_file_mb=512  
llite.*.max_read_ahead_whole_mb=256
```

Command

mlpstorage command	Flags	DLIO params
closed training unet3d run file	--num-accelerators 16 --accelerator-type b200 --client-host-memory-in-gb 768 --num-client-hosts 16	reader.odirect=True reader.read_threads=96 dataset.num_files_train=450002

Run timestamps: 20260723_214126, 20260723_224430, 20260723_234733, 20260724_005036, 20260724_015539, 20260724_025943

Striping

```
lfs setstripe -c -1 -S 4M /mlperf-storage
```

5.15 AMLFS-Durable-Premium-500-40TiB-3-Standard_D192ds_v6

- Tier / capacity: 500 MB/s/TiB, 40 TiB provisioned
- Clients: 3 × Standard_D192ds_v6 (192 vCPU, 768 GiB RAM per SKU; Azure HPC Ubuntu 24.04, Lustre client 2.16.1)

Training, UNet3D

Client tunings (lctl set_param)

```
mdc.*.max_mod_rpcs_in_flight=96
osc.*.max_rpcs_in_flight=128
llite.*.max_read_ahead_mb=2048
llite.*.max_read_ahead_per_file_mb=512
llite.*.max_read_ahead_whole_mb=256
```

Striping

```
lfs setstripe -c -1 -S 4M /mlperf-storage
```

Command

mlpstorage command	Flags	DLIO params
closed training unet3d run file	--num-accelerators 3 --accelerator-type b200 --client-host-memory-in-gb 768 --num-client-hosts 3	reader.odirect=True reader.read_threads=96 dataset.num_files_train=84375

Run timestamps: 20260721_180647, 20260721_190748, 20260721_200852, 20260721_210954, 20260721_221057, 20260721_231200

5.16 AMLFS-Durable-Premium-20-25632TiB-70-Standard_D192ds_v6

- Tier / capacity: 20 MB/s/TiB, 25632TiB provisioned
- Clients: 70 × Standard_D192ds_v6 (192 vCPU, 768 GiB RAM per SKU; Azure HPC Ubuntu 24.04, Lustre client 2.16.1)

Training, UNet3D

Client tunings (lctl set_param)

```
mdc.*.max_mod_rpcs_in_flight=96
osc.*.max_rpcs_in_flight=128
llite.*.max_read_ahead_mb=2048
llite.*.max_read_ahead_per_file_mb=512
llite.*.max_read_ahead_whole_mb=256
```

Striping

```
lfs setstripe -c 10 -S 4M /mlperf-storage
```

Command

mlpstorage command	Flags	DLIO params
closed training unet3d run file	--num-accelerators 70 --accelerator-type b200 --client-host-memory-in-gb 768 --num-client-hosts 70	reader.odirect=True reader.read_threads=96 dataset.num_files_train=1968762

Run timestamps: 20260724_133231, 20260724_143806, 20260724_154411, 20260724_165018, 20260724_175626, 20260724_190328

6. Tuning Reference

The following describes each tuning applied across the runs and the rationale determined during benchmark investigation.

Parameter	Purpose / rationale
mdc.*.max_mod_rpcs_in_flight	Maximum number of modifying metadata RPCs a client may have in flight to each MDT. Raised to 96 to increase metadata write/create concurrency for the many-small-file UNet3D and checkpoint workloads.
osc.*.max_rpcs_in_flight	Maximum concurrent read/write RPCs per OSC (object storage client) to each OST. Raised to 128 to drive higher aggregate throughput and keep the OSSes busy.
llite.*.max_read_ahead_mb	Total client-side read-ahead buffer size. Increased (2 GiB for training, 8 GiB for checkpoint reads) to prefetch large sequential reads.
llite.*.max_read_ahead_per_file_mb	Per-file read-ahead cap. Increased so large individual files (samples / checkpoint shards) prefetch aggressively.
llite.*.hybrid_io / hybrid_io_read_threshold_bytes	Enables the Lustre hybrid I/O read paths; only reads above the 1 GiB threshold take the optimized bulk path, so excluding checkpoint reads. This is used to accelerate large llama3 checkpoint writes, but not reads.
llite.*.max_read_ahead_whole_mb	Threshold below which an entire file is read ahead in one shot; benefits many-small-file training reads.
llite.*.max_cached_mb	Client page-cache size dedicated to Lustre. Set to 32 GB on the E104is v5 checkpoint clients to avoid too long cache flush time between read and writes.
lfs setstripe -c <n> -S <size>	File-system striping: stripe count (number of OSTs a file is spread over, -1 = all) and stripe size. Tuned per workload to balance parallelism and metadata cost.
drop_caches + ldlm.namespaces.*osc*.lru_size=clear	Between checkpoint write and read phases the OS page cache and Lustre DLM lock LRU are flushed to guarantee cold-cache read measurements.